BCS-065	Linux Administration	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	Install **Linux** from a network install server, setup network install servers, and perform automated installs.
CO2	Manage system and services startup and shutdown.
CO3	Select and use system **administration** tools when appropriate.
CO4	Configure and manage printers.
CO5	Use packaging tools to create, install, and de-install packages.

Detailed Syllabus

MODULE-I

System performance-Security-Service Access Control-Network Resource Access Control in Redhat Linux- Goals, Security Domains, System Faults and Analysis methods, Benefits of maintaining system state, Networking-data storage and processing resource concerns, Log file analysis, Understanding service management, Service configuration resource, Implement access control, Selinux overview, SELinux management, IP and IP routing, Compare IPV4 and IPV6, NAT.

MODULE-II

Organizing network system, NFS, SAMBA, Web services in Redhat Linux Host name resolution, Verification of DNS server operation, BIND DNS configuration, DHCP overview and configuration, Describe FTP service, Network file sharing, Samba service, client tool with each service, features of Apache HTTP server, Configure important Apache parameters, pre-directory configuration, CGI use with Apache, identify key modules, introduction to proxy web services.

MODULE-III

Mail services, Securing data, account management with Redhat Linux Understanding email operation, Basic configuration of mail server, Configuring proc mail- dovecot, Debug email services, Fundamentals of encryption protocols, encryption with Redhat linux, configure encryption services for common networking protocols, basic of authentication.

Text Books:-

1. Linux Bible By: Christopher Negus- Wiley Publishing, Inc, 2010
2. Redhat Linux Networking and System Admin BY: Terry Collings and Kurt Wall-M&T Books, 2009

Reference Book:-

1. UNIX and Linux System Administration Handbook (4th Edition), Evi Nemeth, Garth Snyder, Trent R.Hein, Ben Whaley, Prentice Hall; 4th edition(July 24, 2010)
2. Linux Administration A Beginners Guide 6/E, Wale Soyinka, McGraw-Hill Osborne Media; 6th edition (February 21, 2012)

Course Outcomes: After the completion of the course the student will be able to:

CO1	Ability to remember and understand factual knowledge relevant to system administration tools and technologies
CO2	Ability to apply the procedures presented in a "how to" document or tutorial to perform a system administration task successfully
CO3	Ability to use available technical references & resources to find responses to specific system administration questions. This skill entails being able to assert the validity & reliability of such sources
CO4	Ability to write how-to documents, white papers, tutorials guiding other system administrators or users step-by-step through system administration tasks
CO5	Ability to review alternative system administration technologies or solutions based on requirements in order to make recommendation on the most suited